Cube-Sum Permutations: 70 Integer Identities and Four Algebraic Templates

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Abstract

Let $P(n)$ count permutations p of $\{1,\dots,n\}$ for which every $i+p(i)$ is a perfect cube. We prove $P(n)>0$ for every $n \geq 173$ from 70 exceptional integer endpoint identities (927 reflected blocks) and four general algebraic block-rearrangement templates giving eight further seeds. An elementary suffix-reflection rule and a cubic-gap inequality extend the construction to all later lengths. An explicit predecessor table explains all remaining finite lengths without a matching search, a large permutation archive, or dependent augmenting paths. The finite case distinctions are *not* eliminated: this is a solver-free, hand-checkable certificate rather than a case-free parameterized construction.

1 Two elementary lemmas

Lemma 1 (Boundary cancellation)**.** *Let $0=b_0<b_1<\cdots<b_t=n$ and let $c_1,\dots,c_t$ be positive integer cubes. Set $L_j=c_j-b_j$ and $R_j=c_j-b_{j-1}$. Assume $1\leq L_j<R_j\leq n+1$. Then $p(i)=c_j-i$ for $b_{j-1}<i\leq b_j$ is a cube-sum permutation if and only if*

$$\boxed{\{1,R_1,\dots,R_t\}_{\text{multi}}=\{n+1,L_1,\dots,L_t\}_{\text{multi}}.}$$

Proof. Each input block maps bijectively onto the integer interval $[L_j,R_j)$. At an integer x, the jump in the total interval-coverage count is the multiplicity of x as a left endpoint minus its multiplicity as a right endpoint. The displayed identity makes that jump 1 at $x=1$, -1 at $x=n+1$, and zero everywhere else. Thus coverage is exactly one at every $x\in[1,n]$. Reversing this argument proves necessity. □

Lemma 2 (Growing cubic suffix)**.** *Suppose $0\leq m<n$, $m+n+1=c$ is a cube, and a cube-sum permutation on $[1,m]$ exists (the $m=0$ permutation is empty). Keep the existing permutation on $[1,m]$ and set $p(i)=c-i$ for $m<i\leq n$. Then p is a cube-sum permutation on $[1,n]$.*

Proof. The new reflection is a bijection from $[m+1,n]$ to itself, and each new sum is c. □

2 Three algebraic block families and 74 exceptional identities

Lemma 3 (Reflected-block rearrangement)**.** *Choose positive input-block lengths $\ell_1,\dots,\ell_t$ of sum n and an arbitrary output order π of these t blocks. Let A_j be the sum of input lengths before block j, and let B_j be the sum of output lengths before block j in order π. If all numbers*

$$C_j=1+A_j+B_j+\ell_j$$

are perfect cubes, then on input block j the reflection $p(i)=C_j-i$ gives a cube-sum permutation of $[1,n]$.

Proof. The input block is $[A_j+1,A_j+\ell_j]$, which maps exactly to the prescribed output block $[B_j+1,B_j+\ell_j]$. Those output blocks partition $[1,n]$ by construction. Each index sum is C_j. □

Family I (four blocks). Let $U<V$ be integer cubes and put

$$a=2U-V-1,\qquad b=2V-3U+1.$$

If $a,b>0$, use input lengths (a,b,a,b) and output order $(3,4,1,2)$. Lemma 3 gives the cubic sums (U,V,U,V) and length $2(V-U)$. The pairs $(U,V)=(64,125),(125,216),(216,343)$ yield lengths $122,182,254$ respectively.

Family II (five blocks). For cubes $X < Y < Z$, define

$$a = 2Y - Z - 1, \quad b = X + 2Z - 4Y + 1, \quad c = Y - X.$$

If $a, b, c > 0$, the lengths (a, b, c, a, c) in output order $(4, 2, 5, 1, 3)$ produce sums (Y, X, Z, Y, Z) and total length $2Y - X - 1$. With $(Y, Z) = (64, 125)$ and $X = 27, 8$, the results are $n = 100, 119$.

Family III (six blocks). For integer cubes X, W, Y, Z, put

$$\begin{aligned} a &= 2Y - Z - 1, & b &= X - 2a - 1, \\ c &= W + 2a - 2X + 1, & d &= Y + X - W - 2a - 1. \end{aligned}$$

If these are positive, choose lengths (a, b, c, d, a, d) and output order $(5, 2, 3, 6, 1, 4)$. The cubic sums are (Y, X, W, Z, Y, Z) and the length is $2Z - 2Y + X - W$. Taking $(Y, Z) = (64, 125)$, the choices $(X, W) = (8, 64), (27, 64), (8, 27)$ give $n = 66, 85, 103$, respectively.

Family IV (eleven blocks, including degeneration). Let X, Y, Z, W be positive cubes and let Q be an integer. Define

$$\begin{aligned} a &= 4Z - 2W - Y - 1, & b &= 4W + X + 2Y - 8Z + 1, \\ c &= 2Z - W - X, & d &= Q + 4W + 2X - 8Z + 1, \\ e &= 3Z - W - X - Q, & f &= 3W + 2Y - 6Z + 1. \end{aligned}$$

Suppose $a, b, c, e, f > 0$ and $d \geq 0$. The input block lengths are

$$(a, b, c, a, c, d, e, a, f, a, e)$$

and their output order is

$$(10, 2, 5, 8, 3, 6, 11, 4, 9, 1, 7).$$

Lemma 3 gives the successive nominal sums

$$(Z, X, Y, Z, Y, Q, W, Z, W, Z, W)$$

and total length $n = 4Z - W - X - Q - 1$. When $d > 0$, assume Q is also a cube. When $d = 0$, delete the sixth (empty) block from both orders: the value of Q then need *not* be a cube, and $n = 3W + X - 4Z$. With $(Y, Z, W) = (64, 125, 216)$ and $X = 8, 27$, taking $Q = 125$ yields lengths $150, 131$; taking $d = 0$ yields lengths $156, 175$. This proves four cases by a single symbolic family, without comparing four numerical endpoint lists.

Lemma 4 (Affine-word rigidity)**.** *Fix a finite list of integer cubes $C_1, \ldots, C_t$. Suppose all breakpoints $b_j(n)$ are affine functions of n with $b_0(n) = 0$ and $b_t(n) = n$. The resulting fixed-cube reflection word can be a valid permutation for at most two distinct values of n.*

Proof. Every legal permutation satisfies the sum identity

$$\sum_{j=1}^{t} C_j\big(b_j(n) - b_{j-1}(n)\big) = \sum_{i=1}^{n}(i + p(i)) = n(n+1).$$

Its left side is affine in n; its right side is a monic quadratic. The equality cannot hold for three distinct values. □

The obstruction concerns *fixed* cube sums and affine endpoints only, not varying cube sums or composition of reflection templates.

In total these *four symbolic identities* give twelve base lengths

$$66, 85, 100, 103, 119, 122, 131, 150, 156, 175, 182, 254$$

without twelve separate endpoint-table checks.

The remaining 70 rows. The appendix records 70 exceptional rows of the form $n\ b_1 : k_1\ \cdots\ b_t : k_t$. The corresponding cubical sums are $c_j = k_j^3$ and the first boundary is $b_0 = 0$. For every row, the boundaries increase, $1 \le L_j < R_j \le n+1$, and the multiset equality in Lemma 1 holds exactly. These are 927 simple integer-interval identities. Exactly 13 listed rows have $n < 173$ and exactly 57 have $173 \le n \le 364$. Together with the twelve algebraic-template lengths, there are 22 lower and 60 upper seeds.

The following two lists give every omitted base length without further matching data. For any tabulated predecessor m under cube c, define $n = c - m - 1$ and apply Lemma 2. The lists are ordered by cube, so every predecessor is either a seed or was obtained in an earlier row.

First: 26 reflected lengths below 173.

cube	already constructed predecessors m
8	0
27	0,7
64	0,7,19,26
125	0,7,19,26,37,44,56
216	44,56,63,66,68,80,82,85,87, 100,103,105

Together with the 13 exceptional lower seeds and the nine algebraic-template lower seeds, these supply 48 distinct lengths below 173.

Second: 49 reflected lengths in $[173, 255] \cup [339, 364]$.

cube	previously constructed predecessors m
216	0,7,19,26,37
343	0,87,98,100,103,105,110,112, 115,116,117,118,119,120,121,122, 123,124,125,126,128,129,130,131, 132,133,135,147,149,150,151,152, 155,156,159
512	147,149,150,151,152,155,156, 159,171

The 57 upper exceptional rows, the three upper algebraic-template lengths and these 49 reflections are exactly the 109 integers in $[173, 255] \cup [339, 364]$. There is no circular dependency: each predecessor m is strictly smaller than its constructed length n.

For example, $211 = 343 - 131 - 1$, $340 = 512 - 171 - 1$, and $342 = 343 - 0 - 1$. The first example replaces a previously necessary-looking 53-segment independent row by one suffix reflection.

Lemma 5 (Complete finite interval)**.** *For every integer $173 \le n \le 364$, a cube-sum permutation of $[1, n]$ exists.*

Proof. The explicit arithmetic rows and the two predecessor tables establish the claim for $[173, 255] \cup [339, 364]$. For each of the 83 remaining lengths $256 \le n \le 338$, take $m = 511 - n \in [173, 255]$. Then $m < n$ and $m + n + 1 = 512 = 8^3$, so Lemma 2 applies. □

3 A purely symbolic infinite tail

Theorem 1. *For every integer $n \ge 173$, there is a permutation p of $[1, n]$ such that $i + p(i)$ is a perfect cube for all indices i.*

Proof. The base interval is Lemma 5. For $n \ge 365$, assume the claim for every smaller integer at least 173, and let k^3 be the least integer cube satisfying $k^3 \ge n + 174$. Thus $m = k^3 - n - 1 \ge 173$.

If $365 \le n \le 555$, then $k = 9$ and $k^3 = 729 \le 2n$. For $n \ge 556$, minimality of k implies $k \ge 10$ and $n \ge (k-1)^3 - 173$. Therefore

$$\begin{aligned} 2n - k^3 &\ge 2(k-1)^3 - 346 - k^3 \\ &= k^3 - 6k^2 + 6k - 348 \\ &\ge 112 > 0. \end{aligned}$$

The final polynomial equals 112 at $k = 10$ and increases thereafter: its first difference is $3k^2 - 9k + 1 > 0$. In either case $k^3 \le 2n$ and hence $m < n$. Apply the induction hypothesis at m and Lemma 2. □

Logical scope. The theorem uses explicit finite arithmetic equalities and short symbolic lemmas. The optional script `verify_70_plus_templates.py` checks the 70 exceptional rows, regenerates all twelve template cases and checks finite closure without matching or other external data. It is not a premise for the proof. A uniform, case-free formula replacing the remaining 70 exceptional rows is *not* claimed.

A The 70 exceptional integer-multiset witnesses

Every token $b : k$ means $p(i) = k^3 - i$ up to endpoint b. For each row, verify the same endpoint balance of Lemma 1.

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